DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

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Lecture 2: Time Value of Money

Compounding · Discounting · Annuities · Bond Pricing

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Today's Agenda

1. Why is money received tomorrow worth less today?
2. Simple vs. compound vs. continuous compounding
3. Discounting many cash flows: NPV
4. Deriving the annuity formula from scratch
5. Application: pricing a coupon bond

Learning Objectives

- Move any cash flow forward (FV) or backward (PV) in time.
- Compute an effective annual rate from a nominal compounded rate.
- Derive the PV of an ordinary annuity, annuity due, and perpetuity.
- Price a coupon bond as “annuity + single cash flow.”

Why is money received tomorrow worth less today?

A Seed Today Beats a Seed Next Year

Offered one seed today, or one seed in a year – any farmer takes it **today**. Plant it now, and in a year you may have an entire plant's worth of seeds. Money behaves the same way: a rupee today can be invested and *grow*. A rupee promised for later cannot start growing until later.

This – money has a growth opportunity – is the seed every formula today grows from.

Two Operations, All Term

Definition: Time Value of Money

A given sum today is worth more than the same sum later, because of its earning capacity, quantified by a rate r.

Compounding

Move a cash flow *forward* in time $\rightarrow$ Future Value (FV).

Discounting

Move a cash flow *backward* in time $\rightarrow$ Present Value (PV).

Inverse operations, same rate r.

Simple vs. Compound Interest

Simple Interest

$$FV_n = P_0(1 + rn)$$

Only the *original* principal earns interest, every period.

Compound Interest

$$FV_n = P_0(1 + r)^n$$

Interest earned is added to principal and *itself* earns interest.

Compounding Is a Snowball

Interest Earning Interest – A Snowball Rolling Downhill

Simple interest: someone attaches a *fixed* scoop of snow at every turn.
Compound interest: the ball itself gets bigger each turn, so each new layer sticks to an ever-larger surface. That's why, over long horizons, compound growth overtakes linear growth so dramatically.

Example: Simple vs. Compound, Side by Side

Rs. 1,00,000 for 20 years at $r = 8\%$ p.a.

- Simple: $FV = 100000(1 + 0.08 \times 20) = \text{Rs. } 2{,}60{,}000$
- Compound: $FV = 100000(1.08)^{20} \approx \text{Rs. } 4{,}66{,}096$

$\sim$**Rs. 2.1 lakh extra** purely from interest-on-interest.

Compounding m Times a Year

$$FV_n = P_0 \left(1 + \frac{r}{m}\right)^{mn}$$

Effective Annual Rate

$$1 + r_{\text{eff}} = \left(1 + \frac{r}{m}\right)^{m}$$

Remark: read the fine print

"3% per month" sounds mild – but $m = 12 \Rightarrow r_{\text{eff}} = (1.03)^{12} - 1 \approx 42.6\%$ per year!

Continuous Compounding

Let $m \to \infty$. Using $\lim_{m\to\infty}(1 + r/m)^m = e^r$:

$$FV_n = P_0\, e^{rn} \qquad\qquad PV = FV_n\, e^{-rn}$$

Why Bother with an Abstract Limit?

No bank actually compounds every instant. But e^{rt} is smooth and differentiable – the natural language for prices that evolve *continuously*. This is the bridge to Geometric Brownian Motion (Week 8) and Black–Scholes (Week 9).

Three Flavours, Side by Side

Compounding	**FV of P_0**	**PV of FV_n**
Simple	$P_0(1+rn)$	$FV_n/(1+rn)$
Compound (m/yr)	$P_0(1+r/m)^{mn}$	$FV_n(1+r/m)^{-mn}$
Continuous	$P_0 e^{rn}$	$FV_n e^{-rn}$

Discounting Many Cash Flows: NPV

Definition: Net Present Value

$$NPV = CF_0 + \sum_{t=1}^{n} \frac{CF_t}{(1+r)^t}$$

NPV: A Common Currency for Time

Comparing Rs. 100 today to Rs. 110 in a year is like comparing rupees to dollars – you must convert first. NPV converts every cash flow into "rupees *today*" – a common currency any two streams can be honestly compared in.

Deriving the Annuity Formula

Ordinary annuity, payment A, n periods, rate r:

$$PV = \frac{A}{1+r} + \frac{A}{(1+r)^2} + \cdots + \frac{A}{(1+r)^n} = A\sum_{t=1}^{n}(1+r)^{-t}$$

A finite geometric series! Sum it (try this yourself) to get:

$$\boxed{PV = A \cdot \frac{1-(1+r)^{-n}}{r}}$$

Annuity Due and Perpetuity

Annuity Family

- **Ordinary**: $PV = A \cdot \dfrac{1-(1+r)^{-n}}{r}$
- **Annuity due**: $PV_{\text{due}} = PV_{\text{ordinary}} \times (1+r)$
- **Perpetuity** ($n \to \infty$): $PV = \dfrac{A}{r}$

Sanity-Checking the Perpetuity Formula

$PV = A/r$ – Too Simple to Be Right?

A fixed deposit earns Rs. 8 every year forever at 8%. How much principal is sitting in it? Exactly Rs. 100 – since Rs. 100 at 8% generates precisely Rs. 8/year, forever, untouched. The perpetuity formula is just "principal $\times$ rate = interest," solved backwards.

Example: Retirement Annuity

Setting up the problem

Want Rs. 50,000/month for 20 years ($n = 240$), monthly rate $r = 0.5\%$.

$$PV = 50000 \times \frac{1 - (1.005)^{-240}}{0.005} \approx \text{Rs. } 69{,}80{,}000$$

This is exactly how insurers / EPFO price annuity products.

Application: Pricing a Coupon Bond

A bond's coupons *are* an annuity; face value is one cash flow.

Bond Pricing Formula

$$P = \underbrace{C \cdot \frac{1 - (1+y)^{-n}}{y}}_{\text{PV of coupon annuity}} + \underbrace{\frac{F}{(1+y)^n}}_{\text{PV of face value}}$$

y = discount rate per period = the bond's yield to maturity.

Worked Example: Pricing the Lecture 1 Bond

Recall: $F = 100$, coupon $C = 7$, $n = 3$ years, $y = 6\%$

$$P = 7 \cdot \frac{1 - (1.06)^{-3}}{0.06} + \frac{100}{(1.06)^3}$$

$$P = 7(2.6730) + 83.962 \approx \mathbf{102.673}$$

$P > F$: coupon (7%) $>$ yield (6%) $\Rightarrow$ investors pay a *premium*.
(Had $y = 8\%$: you'd find $P < F$, a *discount* – confirming Lecture 1's inverse relationship.)

Reading the Formula Both Ways

Remark

Left-to-right: given yield y, find price P. Right-to-left: given market price P, solve for the **implied yield** y – this *is* yield to maturity.

y cannot generally be isolated algebraically $\rightarrow$ needs numerical root-finding (Newton–Raphson).

Same trick, later: implied volatility from option prices (Week 11).

Key Takeaways

Key Takeaway

- Money has time value: it can be productively invested *starting now*.
- Compounding moves cash forward; discounting moves it back – same rate r.
- Simple grows linearly; compound grows geometrically; continuous (e^{rt}) is the smooth limit we'll need for stochastic models.
- Present values are additive across time – this is why NPV, and annuity pricing, work.
- A bond = annuity (coupons) + single cash flow (face value).

Check Your Understanding

Try these before next lecture

1. Derive the ordinary annuity formula yourself from the geometric series.
2. Bank offers 10% p.a., compounded quarterly. Find r_{eff}.
3. Show algebraically why $P = F$ exactly when coupon rate $=$ yield.
4. A growing perpetuity: $PV = A/(r - g)$. Why does growth *increase* price?

Looking Ahead

So far: *one* discount rate r for every maturity.

But in reality, the 1-year rate $\neq$ the 10-year rate.

The Yield Curve – Lecture 3

...and you'll build one from real data in this week's Lab.